A0/A1 fine-aim sensor mount: design summary

MQ Sentinel · ENGG2000 V3 · Drawing MQS-V3-SNS-001 Rev A (draft) · 2026-10-25 · Prepared by Y. Cui

> Draft for design review. All dimensions come from calculation (TSSP58P38 datasheet, Appendix 1 of the brief, and a 2D ray trace). Nothing has been bench-tested yet. Units mm, tolerance ±0.2 unless stated.

1. What we are asking the design team

Please check whether this mount can be made and fitted as drawn, and tell us what needs to change. The specific questions are in section 9.

2. Job of the part

Two TSSP58P38 IR receivers (A0 left, A1 right) sit at the front of the satellite, on the laser's azimuth. The code compares their readings, e = (L−R)/(L+R), to turn the satellite until the laser points at the active beacon. To score 10 points, the laser must stay inside the 8 cm bullseye at 1 m, which is ±2.3°, for 2 seconds.

The mount must make that comparison change sharply near the centre. It does this with a T-shaped divider (the "T-insert") between the two sensors. Everything has to stay inside the Ø160 pipe.

3. Constraints (brief, Appendix 1)

- Pipe: R80 outer, R76 inner (4 mm wall), 170 tall. Nothing may protrude past R80. Window cuts in the pipe are allowed (confirmed).
- Sensor ring: 40 mm below the top of the body. The IR transmitter sits at this height on each target, and the bullseye at the laser's height, so only azimuth (horizontal) alignment matters.
- Laser pointer: 150 mm below the sensor ring.
- Targets about 1 m away. The part must disassemble with the rest of the satellite in under 10 minutes.

4. Design concept

- **Sensors tilted 20° outward** (40° between their axes), front faces 5 mm either side of the centre line. The tilt keeps the sensors below saturation at 1 m; this was measured on an earlier bench test.
- **T-insert** on the centre line. A thin **stem** stops light crossing over the centre. A front **bar**, 20 mm ahead of the lenses, covers a thin strip of each lens's inner edge. When the beacon moves 1°, the bar's shadow slides about 0.34 mm across the lenses: one sensor gains light while the other loses it.
- **Window frame**, 26 × 10 mm clear, limits the outer field of view and blocks floor, ceiling and side reflections.
- Without the T-insert, both lenses are 100% lit anywhere within ±10° and the comparison carries no information. With it, both are 75% lit at centre, and at 2.3° they read 47% and 97%.

5. Key dimensions

Feature	Value	Note
Housing envelope	30 × 14 × 29.5	W × H × radial depth
Depth inside the pipe	25.5	From the inner wall (R76) inward; 20.5 for the L 15 variant
Window, clear	26 × 10	Printed frame sets the field of view
Pipe window cut	30.4 × 14.4	Centred 40 below pipe top, on the laser azimuth
Sensor pitch	10	Front-face centres ±5 from Datum A
Sensor tilt	20° outward	Each side
Sensor pocket	5.2 × 7.2 × 5.0	TSSP58P38 body 5.0 × 6.95 × 4.8
Stem	3 × 10 × 19	Runs from 1 to 20 in front of the lens plane
Front bar W × H × T	8 × 10 × 2	Back face 20 in front of the lens plane (= L)
Bar overlap on each lens at 0°	≈ 0.9	About 25% of the lens; the most critical feature
Front face	R80	Cylindrical, flush with the pipe outside

6. Parts

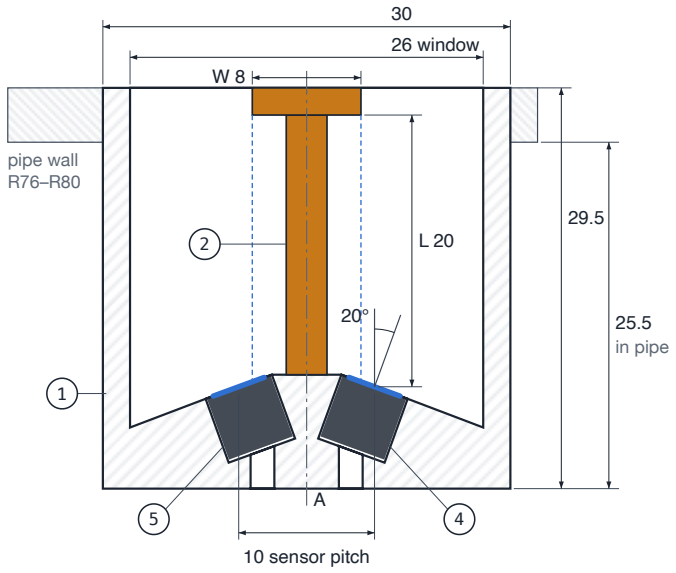
#	Part	Size	Qty
1	Housing: floor, side walls, rear block with 2 sensor pockets and lead slots, window nose	30 × 14 × 29.5	1
2	T-insert (stem + bar), slides down into a 3.2 slot, swappable	stem 3 × 10 × 19, bar W × 10 × 2	3 (W 7, 8, 9)
3	Top cover, holds Part 2; 2 × M2 into the side walls	30 × 2 × 29.5	1
4, 5	TSSP58P38 receivers A1 (right) and A0 (left)	5.0 × 6.95 × 4.8	2
6	Supplied PVC pipe, window cut	30.4 × 14.4	–

Material: PLA, matte black, 100% infill around the pockets and the T-insert. No gloss on any inner surface. Part 1 mounts to the internal chassis by its rear face, not to the pipe.

7. CAD drawing (draft)

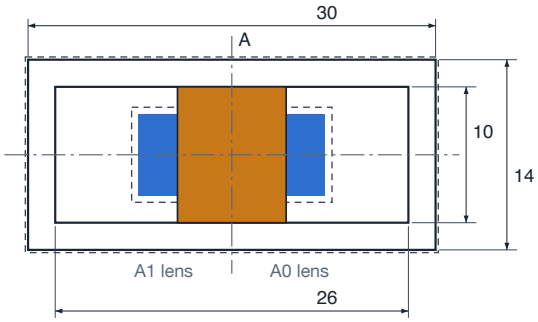
SECTION A–A · TOP, CUT AT LENS HEIGHT · 8:1

Target (beacon / bullseye) is toward the top of this view



VIEW F · FROM THE TARGET SIDE · 8:1

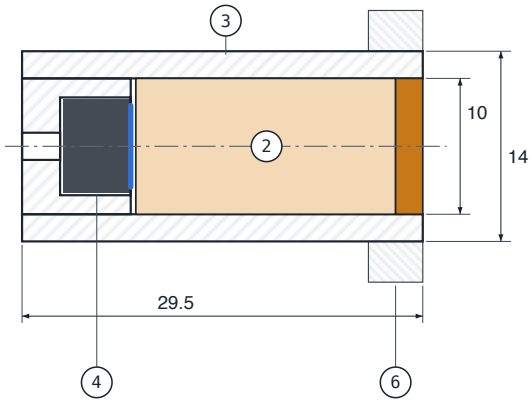
Seen from the target, A1 (right sensor) appears on the viewer's left



Dashed outer rectangle: pipe window cut 30.4 × 14.4

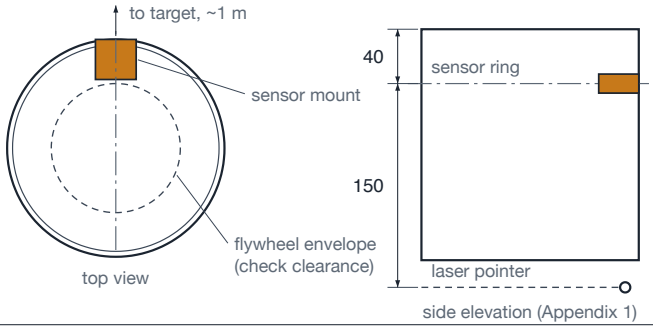
SECTION B–B · SIDE, CUT THROUGH A1 · 8:1

Target is to the right of this view



Light area = stem of Part 2 seen behind the cut; dark amber = front bar. Lens front plane is Z = 0; bar rear face is Z = 20.

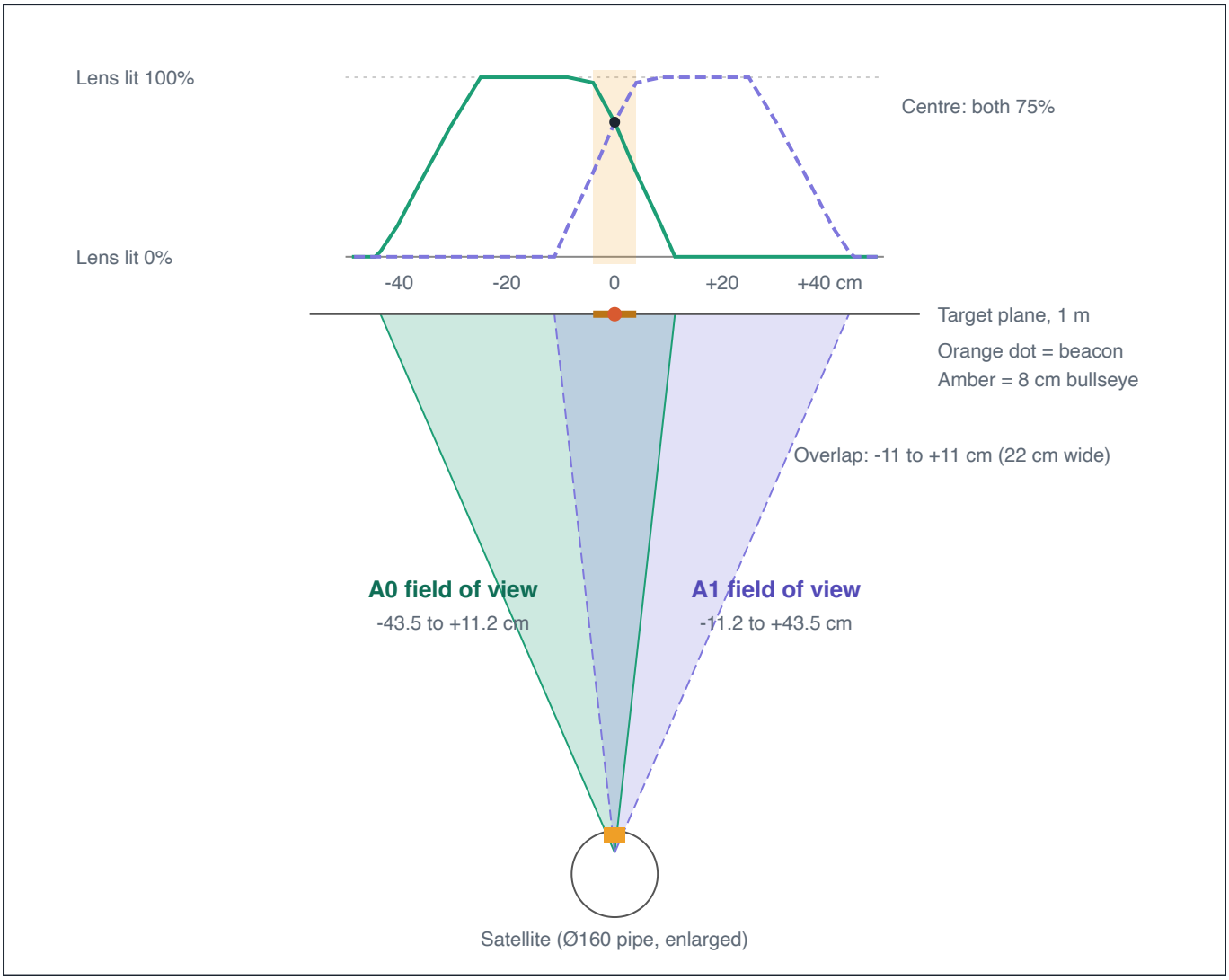
LOCATION IN THE BODY · NOT TO SCALE



Section A–A (top, at lens height), View F (from the target), Section B–B (side, through A1), and location in the body. Hatched = cut material; amber = Part 2 (T-insert); blue = lens; dash-dot A = laser azimuth plane (Datum A).

8. Field of view

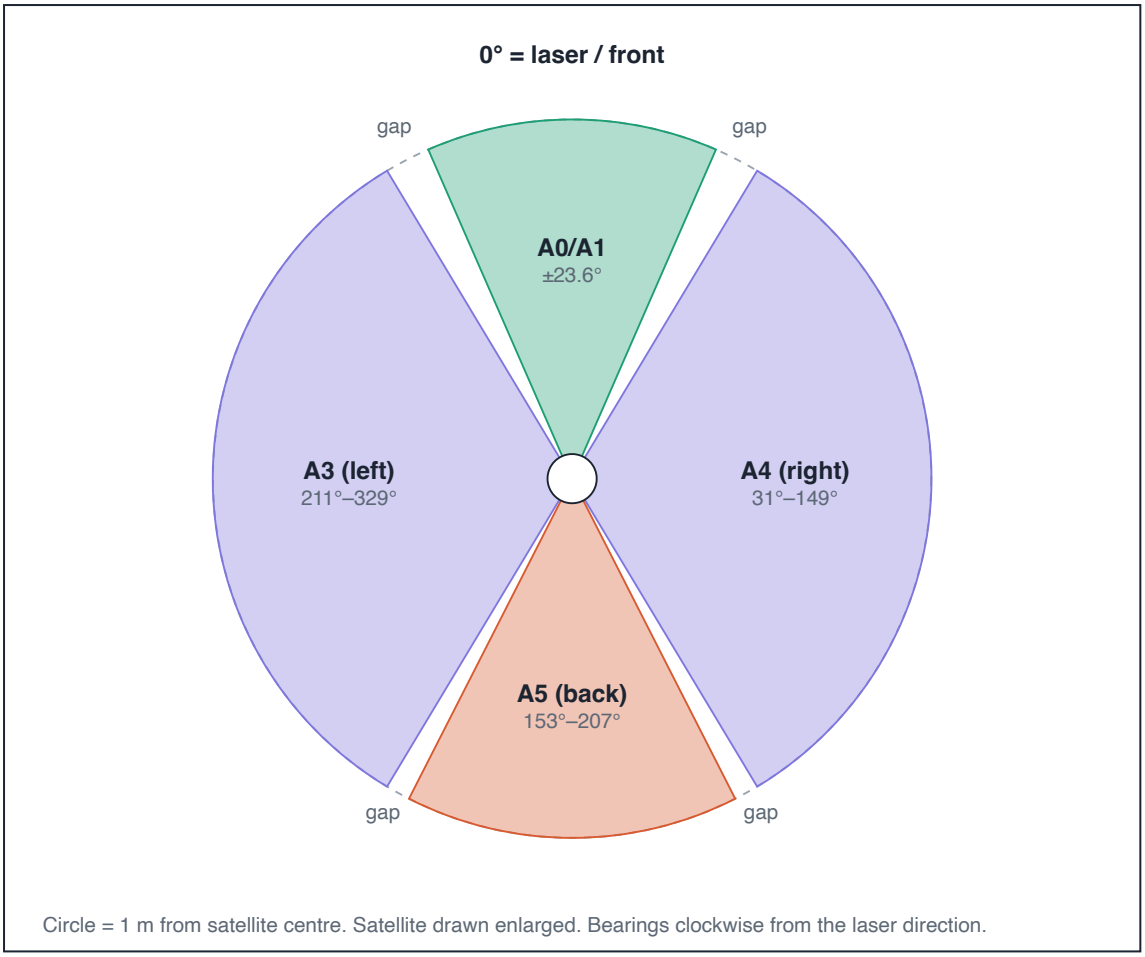
8.1 A0/A1 at 1 m, beacon dead ahead



Ray-traced with the T-insert (W 8, L 20, 20° tilt, 26 mm window). The two views overlap on purpose: the 22 cm overlap (±6.4°) is the zone where L and R are compared for fine aim.

Beacon at 1 m	A0 lens lit	A1 lens lit	Code does
Centre	75%	75%	L ≈ R: hold and fire
4 cm off (2.3°, bullseye edge)	47%	97%	Turn toward the brighter side, slowly
8.7 cm off (5°)	17%	100%	Turn faster
Beyond ±6.4°	One sensor dark		Fixed fast turn toward the lit one
Beyond ±23.6°	Both dark		Hand over to A3–A5

8.2 All sensors around the satellite (A3–A5 for the coarse direction)



Circle = 1 m from satellite centre. Satellite drawn enlarged. Bearings clockwise from the laser direction.

Sensor	Bearing	Lens recess	Window (arc chord)	Fully clear	Cut-off
A0/A1	0°	22 (to R80)	30	±6.4° overlap	±23.6°
A4 / A3	+90° / −90°	13	37.3	36°–144°	31°–149°
A5	180°	20	17.4	162°–198°	153°–207°

- **No physical conflict:** the nearest windows are about 65° of arc apart, about 90 mm on the outside surface.
- **No field-of-view overlap:** there are small gaps between cut-offs, about 7° between front and side and 4° between side and back. These are blind bands; the code handles them with a search step.
- **Nothing blocks the view:** everything is inside the pipe, and a ray leaving a convex cylinder cannot hit it again.
- Each window's edge is a printed knife edge flush with R80. The pipe cut is only clearance. A ±0.5 mm window error moves a side cut-off by about ±0.7°.
- A5 recess options (cut-off fixed at ±27°): 16 mm gives fully clear ±15.8°, 20 mm gives ±18.1° (recommended), 26 mm gives ±20.2°.

9. Questions for the design team

#	Question
1	Is there 25.5 mm of radial space inside R76 at the sensor ring for A0/A1? Or only 20.5 mm (L 15 variant)? Where does the flywheel sit at that height?
2	Is there room for A3/A4 (13 mm recess) and A5 (20 mm recess) at the same height?
3	Can the 0.9 mm bar overlap be printed and held to ±0.2? This sets the centre offset. Can the 3.2 slot keep the T-insert from rocking?
4	Can the nose and bar front faces be printed as an R80 cylinder? If not, the face must be set back about 1.7 mm at the centre so the corners stay inside R80.
5	How should Part 1 mount to the internal chassis while keeping the centre line on the laser azimuth within ±0.5 mm and ±0.5° yaw?
6	Where do the leads and wiring run from the rear lead slots to the Nano?
7	Does this still disassemble with the rest of the satellite in under 10 minutes?

10. Status and risks

- A cardboard version of the T-insert is being built to bench-test the readings before printing.
- The TSSP's pulse width is roughly logarithmic in light level, so real readings change less than the lit percentages above. The bench test sets the final W, L and thresholds.
- If the T-insert doesn't give a strong enough change, the fallback is an edge-finding method in software, using the same hardware.